Ashh Research Paper: Attention and Retrieval Augmented Generation
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Abstract: This paper presents modern methods in grounded question answering.
Retrieval Augmented Generation (RAG) significantly reduces model hallucinations by anchoring responses in verified document chunks.
The system chunks incoming text into approximately 800 characters with 120 character overlaps.
Dense vector embeddings index every chunk in a vector store like ChromaDB.

Section 2: Vector Space Search & Cosine Similarity
In ChromaDB, chunk embeddings are mapped onto a high-dimensional manifold.
When a query arrives, top-k chunks are selected based on nearest neighbor metrics.
Top-k defaults to 5 chunks, but can be configured between 1 and 10 chunks depending on context budget.
Document scoping enables users to constrain search strictly to specific files.

Section 3: Verifiable Numbered Source Citations
Grounded models cite sources using bracketed tokens such as [1] and [2].
Every statement can be verified against the source text and exact page number.
When no documents are loaded, the system gracefully advises the user to upload reference material.
Offline extractive mode ensures uninterrupted operation even without cloud LLM keys.